University of Asia Pacific

Department of Computer Science and Engineering

CSE 316: Microprocessors and Microcontrollers Lab

LAB REPORT

Experiment Number: 04

Experiment Title: Touch Sensor Based Door Lock with Buzzer and Servo Motor

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1. Experiment Name

Touch Sensor Based Door Lock with Buzzer and Servo Motor

2. Objective

To design and implement a touch sensor–based electronic door lock system that uses a pushbutton as the touch input, a servo motor for the locking/unlocking mechanism, and a piezo buzzer for auditory feedback. The objective is to simulate a smart, low-cost door locking solution that provides security while demonstrating Arduino-based control of actuators and sensors.

3. Apparatus / Hardware & Software Requirements

Hardware
-1 × Arduino Uno R3 -1 × Servo Motor -1 × Piezo Buzzer -1 × Pushbutton -1 × Breadboard -1 × Resistor (1 kΩ) -1 × USB Cable withpower supply -Jumper Wires

Software
-Arduino IDE -Tinker CAD

4. Circuit Diagram / Schematic :

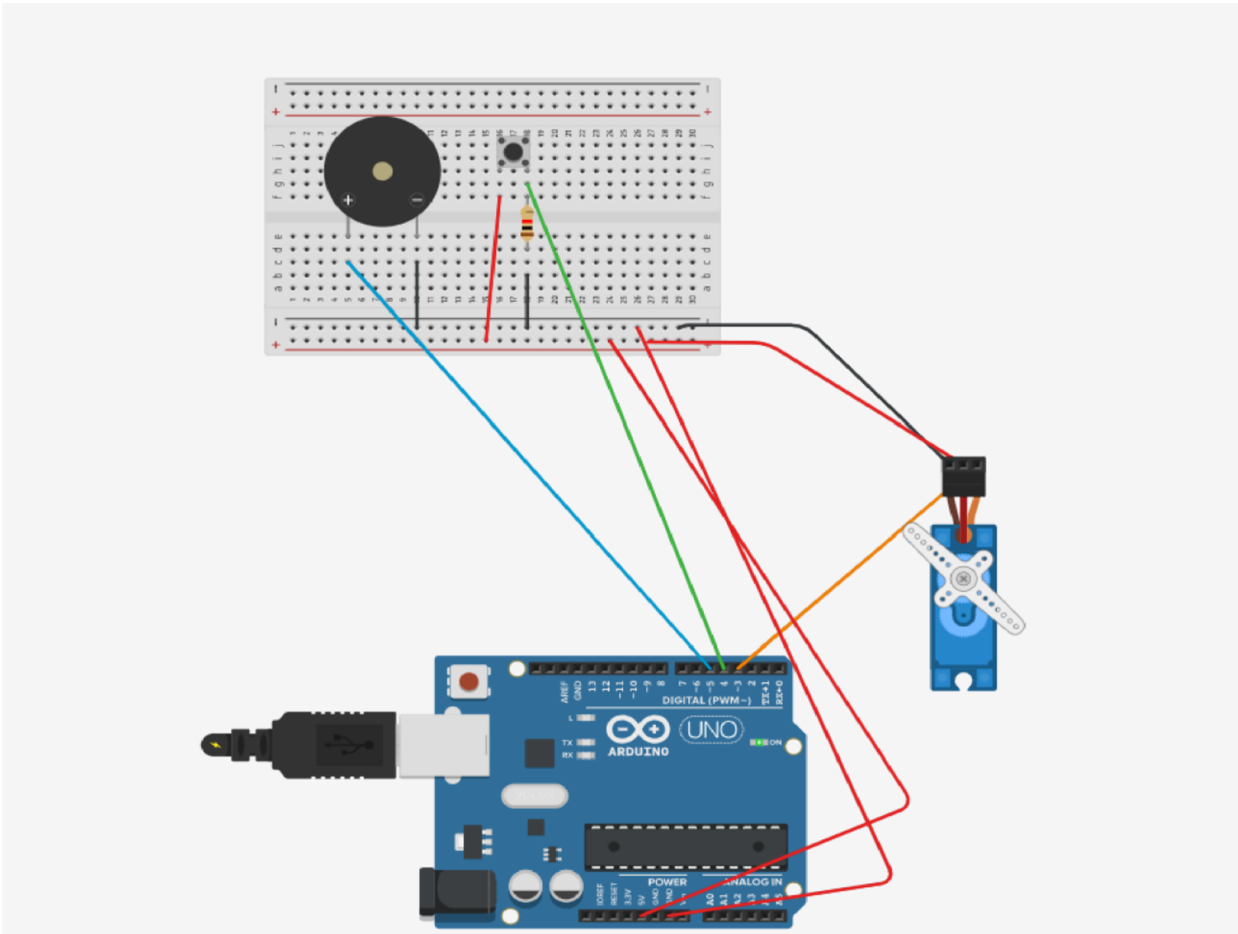

Figure 1: Circuit diagram of a Touch Sensor Based Door Lock with Buzzer and Servo Motor

5. Code / Assembly Program:

```
#include <Servo.h>
bool doorlocked = true;               // Initially locked
Servo myServo;
const int touchpin = 4;
const int buzzpin = 5;
void setup() {
  pinMode(touchpin, INPUT);
  pinMode(buzzpin, OUTPUT);
  myServo.attach(3);
  myServo.write(45);                  // Start locked position
  Serial.begin(9600);
}
void loop() {
  int touchState = digitalRead(touchpin);
  if (touchState == HIGH) {
    if (doorlocked) {                 // Unlock door
      myServo.write(90);
      tone(buzzpin, 400, 200);
      delay(200);                     // Short beep 1
      tone(buzzpin, 600, 300);
      delay(300);                     //Short beep 2
      noTone(buzzpin);
      doorlocked = false;
    }
    else {                            // Lock door
      myServo.write(45);              // Rotate to lock position
      tone(buzzpin, 600, 200);        // short beep 1
      delay(200);
      tone(buzzpin, 400, 300);        // short beep 2
      delay(300);
```

```
        noTone(buzzpin);
        doorlocked = true;
      }
      delay(800);   // Prevent multiple triggers
    }
  }
```

6. Output / Observations:

1. The pushbutton was connected to Arduino digital pin 4 with a 1 kΩ pull-down resistor
2. The servo motor was connected with Arduino digital pin 3
3. The piezo buzzer was connected to Arduino digital pin 5
4. The circuit was powered via the Arduino Uno's USB supply, and all components shared a common ground.

When executed in Tinkercad, pressing the pushbutton toggled the servo position (45° when locked, 90° when unlocked) and activated distinct buzzer tones:

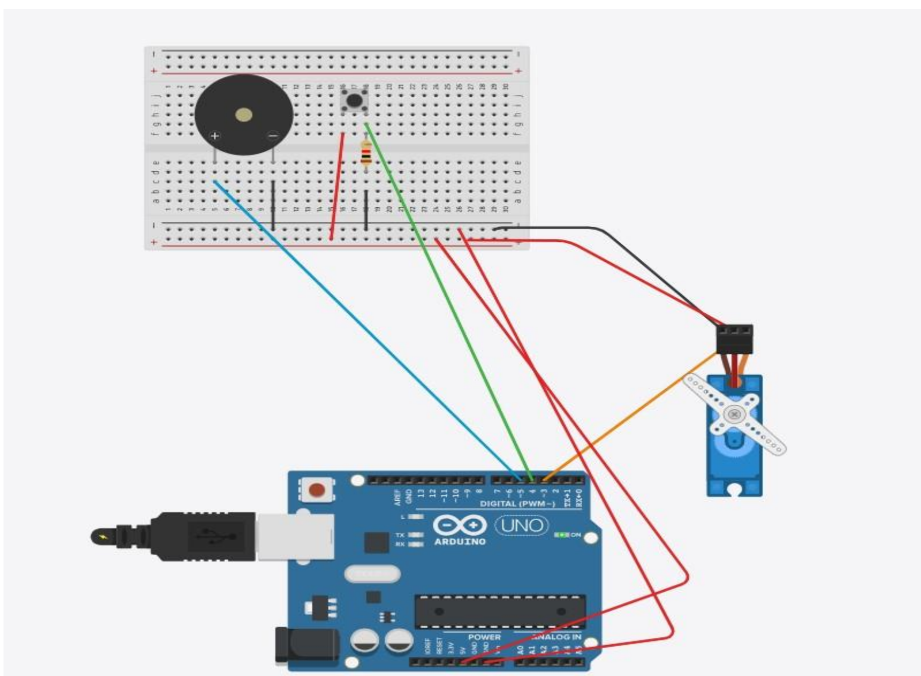

Figure 2: **State of servo motor (45°) and piezo buzzer when door is locked**

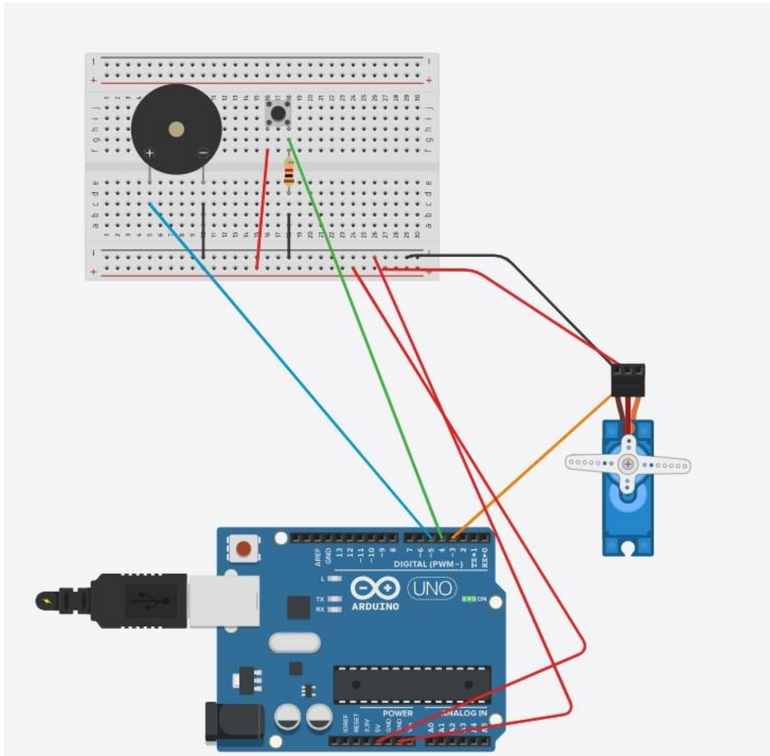

Figure 3 : State of servo motor (90°) and piezo buzzer when door is unlocked

7. Result:

The door lock system operated according to the intended design. It effectively demonstrated a functional touch-controlled locking mechanism. The servo motor correctly illustrated both the locked and unlocked positions, while the buzzer delivered clear audio signals to confirm system actions. The Arduino handled the input signals and managed the output devices smoothly. In general, the implementation confirmed that the programmed logic worked correctly and that the system performed its expected functions.

8. Conclusion:

This project shows how microcontrollers can combine sensing, decision-making, and device control in a real-world application. The touch sensor served as an easy and user-friendly input method, while the servo motor acted as a model for the locking mechanism. The buzzer also improved the user experience by providing sound signals that indicate different system states. Even though the project was implemented on a small scale, the core design concepts can be applied to more complex security systems. In the future, adding features such as capacitive touch sensors or a keypad with PIN verification could further strengthen system security. Overall, the project achieved its goals and provided practical experience in integrating different components within an embedded system.